Solutions to 2013 Power Question

1. a. Factoring, $6 = 2 \cdot 3^1$, $16 = 16 \cdot 3^0$, and $72 = 8 \cdot 3^2$, so $d(6) = 1/3$, $d(16) = 1$, and $d(72) = 1/9$.

   b. If $n = 3^k m$ where $3 \nmid m$, then $d(n) = 1/3^k$. So the smallest values of $d(n)$ occur when k is largest. The largest power of 3 less than 100 is $3^4 = 81$, so $d(81) = 1/3^4 = 1/81$ is minimal.

   c. If $d(n, 17) = 1/9$, then $9 \mid (n - 17)$ and $27 \nmid (n - 17)$. So $n = 17 + 9k$ where $k \in \mathbb{Z}$ is not divisible by 3. For $k = -1$, 1, 2, 4, this formula yields $n = 8$, 26, 35, and 53, respectively.

   d. There are many possible answers; the simplest is $h_n = 17 + 3^n$, yielding the sequence $\{20, 26, 44, 98, \ldots\}$.

2. a. Because $d(6) = 1/3$, $\mathcal{N}(6)$ is the set of all houses n such that $d(n) = 1/3$, equivalently, where $n = 3k$ and k is not divisible by 3. So $\mathcal{N}(6) = \{3, 6, 12, 15, 21, 24, \ldots\}$.

   b. Using similar logic to 2a, $\mathcal{N}(n) = \{m \in \mathbb{Z}^+ \mid 3 \nmid m\}$. So $\mathcal{N}(n) = \{1, 2, 4, 5, 7, 8, \ldots\}$.

   c. Here, $\mathcal{N}(n) = \{m \mid m = 27k, \text{ where } 3 \nmid k\}$. The ten smallest elements of $\mathcal{N}(n)$ are 27, 54, 108, 135, 189, 216, 270, 297, 351, and 378.

3. a. The statement is false. $d(n, m) < 1$ if $3 \mid (n - m)$, so for example, if $n = 2$ and $m = 5$.

   b. If $d(m) = d(n) = 1/3^k$, rewrite $m = 3^k m'$ and $n = 3^k n'$ where m' and n' are not divisible by 3. Then $m - n = 3^k(m' - n')$. If $3 \mid (m' - n')$, then $d(m, n) < 1/3^k$. If $3 \nmid (m' - n')$, then $d(m, n) = 1/3^k$. So the set of possible values of $d(m, n)$ is $\{3^{-q} \mid q \geq k, q \in \mathbb{Z}^+\}$.

   c. Because $d(17, m) = 1/81$, $17 - m = 81l$, where $l \in \mathbb{Z}$ and $3 \nmid l$. So $m = 17 - 81l$ and $m - 16 = 1 - 81l$. Hence $3 \nmid m - 16$, and $d(m, 16) = d(m - 16) = 1$.

4. a. Because $d(17, 34) = d(17) = 1$, $d(34, 51) = d(17) = 1$, and $d(17, 51) = d(34) = 1$, it follows that $d(17, 51) \leq d(17, 34) + d(34, 51)$. Because $d(17, 68) = d(51) = 1/3$, $d(68, 95) = d(27) = 1/27$, and $d(17, 95) = d(78) = 1/3$, it follows that $d(17, 95) \leq d(17, 68) + d(68, 95)$.

   b. Because $d(17, 68) = 1/3$, the condition implies that $3 \mid (c - 17)$ but $9 \nmid (c - 17)$. Thus either $c - 17 = 9k + 3$ or $c - 17 = 9k + 6$ for some $k \in \mathbb{Z}$. Solving for c yields $c = 9k + 20$ or $c = 9k + 23$, equivalently $c = 9k + 2$ or $c = 9k + 5$, $k \in \mathbb{Z}$, $k \geq 0$.

5. a. Write $a = 3^\alpha a_0$ and $b = 3^\beta b_0$, where $3 \nmid a_0$ and $3 \nmid b_0$. First consider the case $\alpha = \beta$. Then $a - b = 3^\alpha(a_0 - b_0)$. In this case, if $3 \mid (a_0 - b_0)$, then $3^\alpha(a_0 - b_0) = 3^\gamma c$, where $3 \nmid c$ and $\alpha < \gamma$; so $d(a - b) = 3^{-\gamma} < 3^{-\alpha} = d(a) = d(b)$. If $3 \nmid (a_0 - b_0)$, then $d(a, b) = 3^{-\alpha} = d(a) = d(b)$. If $\alpha \neq \beta$, suppose, without loss of generality, that $\alpha < \beta$ (so that $d(a) > d(b)$). Then $a - b = 3^\alpha(a_0 - 3^{\beta-\alpha}b_0)$. In this second factor, notice that the second term, $3^{\beta-\alpha}b_0$ is divisible by 3 but the first term a_0 is not, so their difference is not divisible by 3. Thus $d(a, b) = 3^{-\alpha} = d(a)$. Therefore $d(a, b) = d(a)$ when $d(a) > d(b)$, and similarly $d(a, b) = d(b)$ when $d(b) > d(a)$. Hence $d(a, b) \leq \max\{d(a), d(b)\}$.

b. Note that $d(a,c) = d(a-c) = d((a-b)-(c-b)) = d(a-b, c-b)$. Then $d(a,c) \leq \max\{d(a,b), d(c,b)\}$ by part 5a, and the fact that $d(c,b) = d(b,c)$.

c. Because $d(x,y) > 0$ for all $x \neq y$, $\max\{d(a,b), d(b,c)\} < d(a,b) + d(b,c)$ whenever $a \neq b \neq c$. Thus $d(a,c) < d(a,b) + d(b,c)$ when a, b, c are all different. If $a = b \neq c$, then $d(a,b) = 0$, $d(a,c) = d(b,c)$, so $d(a,c) \leq d(a,b) + d(b,c)$. And if $a = c \neq b$, then $d(a,c) = 0$ while $d(a,b) = d(b,c) > 0$, so $d(a,c) < d(a,b) + d(b,c)$.

d. The foregoing shows that all ARMLopolitan triangles are isosceles! Examining the proof in 5a, note that if $d(a) \neq d(b)$, then $d(a,b) = \max\{d(a), d(b)\}$. Applying that observation to the proof in 5b, if $d(a,b) \neq d(b,c)$, then $d(a,c) = \max\{d(a,b), d(b,c)\}$. So either $d(a,b) = d(b,c)$ or, if not, then either $d(a,c) = d(a,b)$ or $d(a,c) = d(b,c)$. Thus all ARMLopolitan triangles are isosceles.

6. a. For example, $d(9) = 1/9$ and $d(9{+}18) = 1/27$. In general, if $n = 9{+}27k$, then $d(n) = 1/9$, but $d(n+18) = d(27+27k) \leq 1/27$.

b. Note that $d(m{+}18, n{+}18) = d((m{+}18){-}(n{+}18)) = d(m{-}n) = d(m,n)$. In particular, if $m = n+1$, then $d(m,n) = 1$, thus $d(m+18, n+18)$ is also 1.

7. a. No, they won't move. If $a \in \mathcal{N}(1)$, then $3 \nmid a$. Thus $3 \nmid (a+18)$, and so $d(a+18) = 1$.

b. Yes, they will be split apart. If $a \in \mathcal{N}(9)$, then $9 \mid a$ but $27 \nmid a$, so $a = 27k+9$ or $a = 27k+18$. Adding 18 has different effects on these different sets of houses: $27k{+}18 \mapsto 27k{+}36$, which is still in $\mathcal{N}(9)$, while $27k{+}9 \mapsto 27k{+}27$, which is in $\mathcal{N}(27r)$ for some $r \in \mathbb{Z}^+$, depending on the exact value of a. For example, $a = 9 \mapsto 27$, while $a = 63 \mapsto 81$, and $d(81) = 3^{-4}$. That is, $d(a+18) \leq 3^{-3}$ whenever $a = 27k+9$, with equality unless $a \equiv 63 \bmod 81$.

c. The question is whether it is possible that $x \notin \mathcal{N}(9)$ but $x + 18 \in \mathcal{N}(9)$. Because $d(9) = 1/9$, there are two cases to consider: $d(x) > 1/9$ and $d(x) < 1/9$. In the first case, $9 \nmid x$ (either x is not divisible by 3, or x is divisible by 3 and not 9). These houses do not move into $\mathcal{N}(9)$: $x \not\equiv 0 \bmod 9 \Rightarrow x + 18 \not\equiv 0 \bmod 9$. On the other hand, if $d(x) < 1/9$, then $27 \mid x$, i.e., $x = 27k$. Then $x + 18 = 27k + 18$ which is divisible by 9 but not by 27, so $d(x+18) = 9$. So in fact **every** house x of ARMLopolis with $d(x) < 1/9$ will move into $\mathcal{N}(9)$!

d. As noted in 7c, all houses in $\mathcal{N}(3^k)$, where $k \geq 3$, will move into $\mathcal{N}(9)$. On the other hand, houses of the form $27k + 9$ will find that their distance from City Hall *decreases*, so each of those now-vacated neighborhoods will be filled by residents of the old $\mathcal{N}(9)$. None of $\mathcal{N}(1)$, $\mathcal{N}(2)$, or $\mathcal{N}(3)$ will be affected, however.

8. a. If $n = 3k+2$, then $d(23,n) = d(3k-21)$, and because $3k - 21 = 3(k-7)$, $d(3k-21) \leq 1/3$. On the other hand, if $d(23,n) \leq 1/3$, then $3 \mid (n-23)$, so $n - 23 = 3k$, and $n = 3k + 23 = 3(k+7) + 2$.

b. If $n = 3k+2$, then $d(32,n) = d(3k-30) \leq 1/3$. Similarly, if $d(32,n) \leq 1/3$, then $3 \mid n - 32 \Rightarrow n = 3k + 32 = 3(k+10) + 2$.

c. Show that $\mathcal{D}_r(x) \subseteq \mathcal{D}_r(z)$ and conversely. Suppose that $y \in \mathcal{D}_r(x)$. Then $d(x, y) \leq r$. Because $d(x, z) = r$, by 5b, $d(y, z) \leq \max\{d(x, y), d(x, z)\} = r$. So $y \in \mathcal{D}_r(z)$. Thus $\mathcal{D}_r(x) \subseteq \mathcal{D}_r(z)$. Similarly, $\mathcal{D}_r(z) \subseteq \mathcal{D}_r(x)$, so the two sets are equal.

9. a. The maximum possible distance $d(34, n_k)$ is $1/3$. This can be proved by induction on k: $d(n_1, 34) \leq 1/3$, and if both $d(n_{k-1}, 34) \leq 1/3$ and $d(n_{k-1}, n_k) \leq 1/3$, then $\max\{d(n_{k-1}, 34), d(n_{k-1}, n_k)\} \leq 1/3$ so by 5b, $d(34, n_k) \leq 1/3$.

b. The sequence is contained in the disk for all rational values of r, by the same logic as in 9a.

10. a. $d(3/5) = 1/3$, $d(5/8) = 1$, and $d(7/18) = 9$.

b. Because $d(4/3) = 1/3$, $\mathcal{N}(4/3) = \{p/q$ in lowest terms such that $3 \mid q$ and $9 \nmid q\}$. Thus the set of all possible (p, q) consists precisely of those ordered pairs $(p'r, q'r)$, where r is any non-zero integer, p' and q' are relatively prime integers, $3 \mid q'$, and $9 \nmid q'$.

c. The houses in this sequence actually get arbitrarily far apart from each other, and so there's no single house which they approach. If $H_n = \frac{1}{0!} + \frac{1}{1!} + \cdots + \frac{1}{n!}$, then $d(H_n, H_{n-1}) = d(1/n!) = 3^k$, where $k = \lfloor n/3 \rfloor + \lfloor n/9 \rfloor + \lfloor n/27 \rfloor \ldots$. In particular, if $n! = 3^k \cdot n_0$, where $k > 0$ and $3 \nmid n_0$, then $d(H_n, H_{n-1}) = 3^k$, which can be made arbitrarily large as n increases.